MAINCRAFTS TECHNOLOGY

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UAV Design & Simulation Internship - Task 4

Autonomous Navigation + GPS Waypoint Mission Planning

Objective

In this task, interns will:

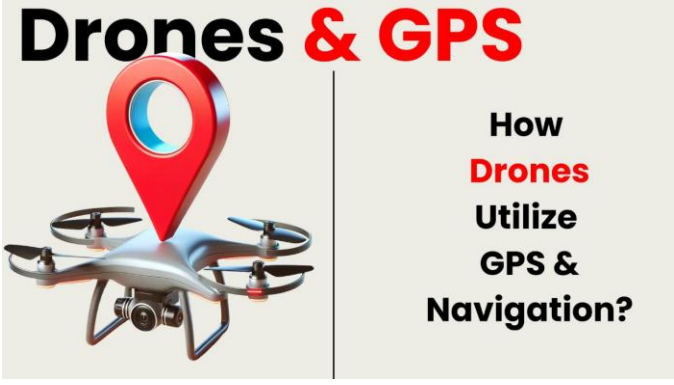
- Understand **GPS-based drone navigation**
- Learn **waypoint mission planning**
- Execute **autonomous flight in simulation (PX4 + Gazebo + QGroundControl)**
- Understand **how real delivery, surveillance, and mapping drones operate**

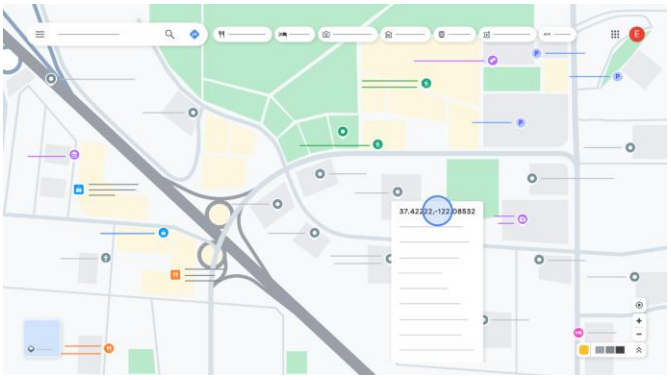
This is a critical industry skill used in:

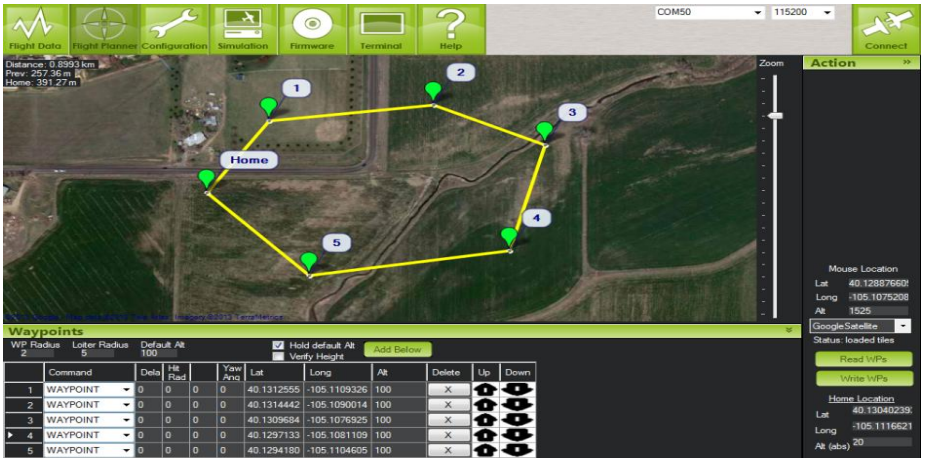
- Drone delivery systems
- Agricultural mapping
- Defense surveillance
- Smart city monitoring

Part 1: Understanding GPS-Based Navigation

What is GPS in UAV?







GPS provides:

- Latitude
- Longitude
- Altitude

Drone uses GPS to:

- Know its position
- Navigate autonomously
- Follow predefined paths

Key Concepts

Intern must understand:

- Latitude & Longitude
- Waypoints (target positions)
- Home position
- Return-to-Launch (RTL)

Deliverable 1:
Document explaining:

- How GPS works in drones
- What are waypoints
- Real-world use cases

Part 2: Waypoint Mission Planning (QGroundControl)

What is a Waypoint Mission?

A sequence of GPS points that a drone follows automatically.

Mission Flow

1. Set home location
2. Add waypoints on map
3. Define altitude
4. Upload mission
5. Start mission

Tasks to Perform

- Open QGroundControl
- Select map view
- Add at least 4–5 waypoints
- Set altitude (e.g., 10–20 meters)
- Save mission

Deliverable 2:

- Screenshot of waypoint map
- List of coordinates used

Part 3: Autonomous Flight Execution

Steps

1. Start PX4 SITL + Gazebo
2. Connect QGroundControl
3. Upload mission
4. Arm drone
5. Start mission

Expected Behavior

- Drone takes off automatically
- Moves between waypoints
- Maintains altitude
- Returns to home (optional)

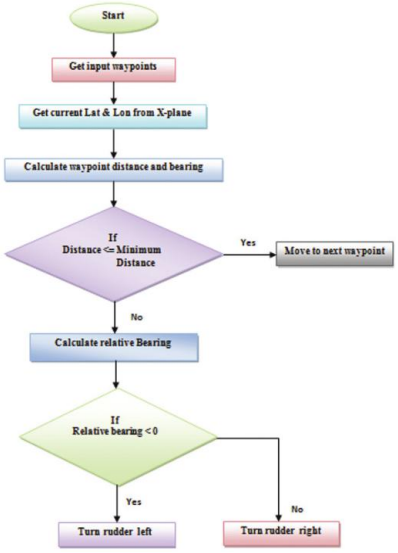
Deliverable 3:

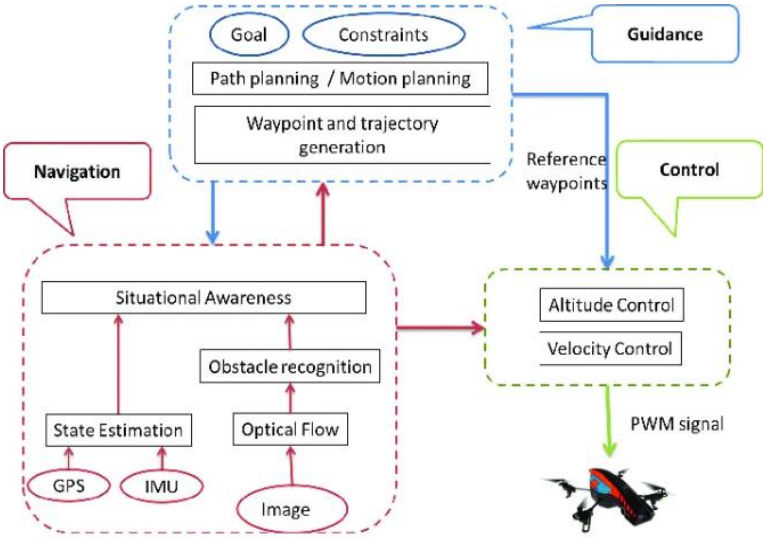
Screen recording showing:

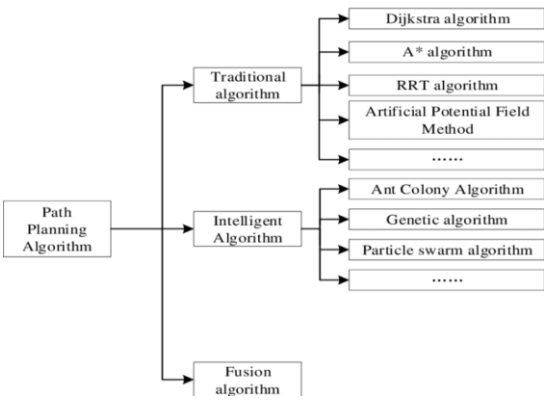
- Full mission execution

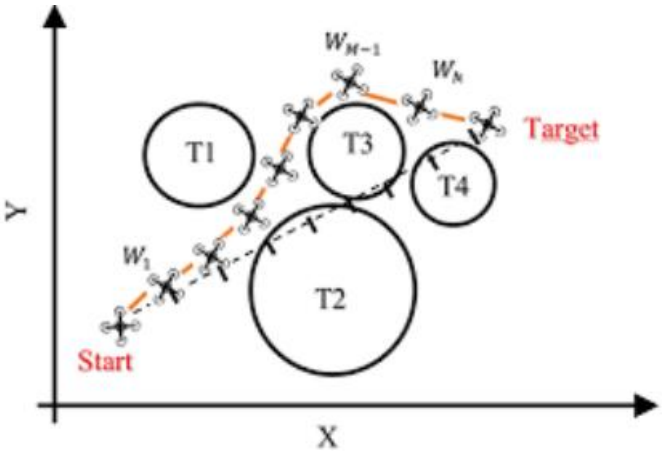
Part 4: Understanding Navigation Logic

Internal Working









Flow:

GPS Data → Position Estimation → Path Planning → PID Control → Motor Output

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Intern must explain:

- How drone moves from one waypoint to another
- Role of PID (from Task 2) in navigation
- How errors are corrected

Deliverable 4:
2–3 page explanation

Part 5: Mini Project (Industry-Level Scenario)

Mission: Area Survey Simulation

Scenario:

- Drone must cover a specific area (like farmland or city block)

Task

- Create 5–6 waypoints forming a path
- Maintain consistent altitude
- Complete full mission
- Return to launch

Expected Output

- Smooth navigation
- Stable altitude
- Accurate waypoint following

Deliverable 5:

- Mission video
- Explanation of navigation behavior

Part 6: Advanced Challenge (Optional)

Add Real-World Complexity

Intern can try:

1. Increase number of waypoints
2. Change altitude between points
3. Simulate disturbance (wind)
4. Observe deviation and correction

Deliverable 6:
Short analysis of system behavior under complexity

Part 7: Industry Connection

Where this is used:

- Delivery drones (Amazon-style logistics)
- Agricultural spraying and mapping
- Defense border surveillance
- Disaster management drones
- Infrastructure inspection

Intern must write:

- Why autonomous navigation is important
- Benefits over manual control

Learning Outcome After Task 4

By end of this task, intern will:

- Plan waypoint missions
- Execute autonomous drone flight
- Understand GPS-based navigation
- Analyze real UAV mission behavior
- Work like industry-level UAV engineer

Below are **100% Free Learning Resources** + **YouTube Links** + **Tools** + **Support** required to complete **Task 4: Autonomous Navigation + GPS Waypoint Mission Planning.**
All resources are aligned with **industry-level UAV navigation systems (PX4, QGroundControl, MAVLink, GPS-based missions)** and follow a **self-learning model.**

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1. GPS & Drone Navigation Fundamentals
A. Basics of GPS (Latitude, Longitude, Altitude)
Khan Academy
YouTube: https://www.youtube.com/@khanacademy
Search Topics:

- Latitude and longitude basics
- How GPS works
- Coordinates on maps

NASA
YouTube: https://www.youtube.com/@NASA
Search:

- How GPS works satellites
- Navigation systems basics

B. UAV Navigation Concepts
MIT OpenCourseWare
YouTube: https://www.youtube.com/@mitocw

Search:

- Autonomous navigation
- Path planning robotics

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2. QGroundControl (Mission Planning Tool)

QGroundControl

Official Website:

https://qgroundcontrol.com

YouTube Learning

Search:

- QGroundControl waypoint mission tutorial
- How to create mission plan QGroundControl
- Upload mission PX4 QGroundControl

What to Learn:

- Map interface
- Adding waypoints
- Setting altitude
- Mission upload

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3. PX4 Autonomous Navigation

PX4 Autopilot

Documentation:

https://docs.px4.io

YouTube Learning

Official Channel:

https://www.youtube.com/@PX4Autopilot

Search:

- PX4 waypoint mission
- PX4 autonomous flight
- PX4 position mode

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4. MAVLink Protocol (Basic Understanding)

MAVLink

Website:

https://mavlink.io

What to Learn:

- Communication between drone and ground station
- How commands are sent

Search:

- MAVLink basics
- Drone telemetry communication

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5. Path Planning Concepts (Important for Industry)

Basic Topics to Search

- What is path planning in robotics
- Waypoint navigation algorithm
- Grid-based path planning
- A* algorithm basics (optional advanced)

YouTube Learning

Steve Brunton

YouTube: https://www.youtube.com/@Eigensteve

Search:

- Robotics path planning
- Autonomous systems

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6. Maps & Coordinate Understanding

Tools

- Google Maps
- OpenStreetMap

Search:

- How to find latitude longitude Google Maps
- How GPS coordinates work

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7. Tools Required

Software (Must Have)

- PX4 SITL (from Task 3)
- Gazebo Simulator
- QGroundControl

Optional Tools

- Mission Planner (ArduPilot alternative)
- Python (for advanced automation later)

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8. Step-by-Step Learning Plan

Day 1-2

- Learn GPS basics
- Understand coordinates and mapping

Day 3-4

- Learn QGroundControl interface
- Practice adding waypoints

Day 5

- Study PX4 autonomous navigation
- Understand position mode

Day 6-7

- Run simulation
- Execute waypoint mission

Day 8-9

- Analyze telemetry
- Observe movement accuracy

Day 10

- Complete mini project (area survey mission)

9. Practice Exercises

Intern should try:

1. Create 3 waypoint mission
2. Increase to 5–6 waypoints
3. Change altitude at each point
4. Enable Return-to-Launch
5. Observe path accuracy

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10. Common Mistakes to Avoid

- Incorrect waypoint placement
- Too low altitude (collision risk)
- Not arming drone properly
- Ignoring GPS lock
- Not checking mission upload

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11. System Requirements

Minimum:

- 8 GB RAM
- Stable internet
- Working PX4 setup

Recommended:

- 16 GB RAM for smooth simulation

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12. Final Submission Checklist

Intern must submit:

- GPS + waypoint explanation document
- Screenshot of mission planning
- Coordinates list
- Screen recording of autonomous flight
- Navigation analysis report